Fluid Universe Model: A Unified Framework for Cosmology, Gravity and Quantum Emergence

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ABSTRACT
The standard Lambda-Cold Dark Matter model and General Relativity face severe fundamental difficulties, including the lack of direct detection of dark matter and dark energy, unavoidable spacetime singularities, and the irreconcilable incompatibility between gravity and quantum mechanics. This paper proposes a complete Fluid Universe Model based on the core hypothesis that spacetime itself is a compressible and viscous cosmic fluid. According to the basic conservation laws of fluid mechanics, this paper derives a closed system of equations consisting of the continuity equation, momentum equation, vorticity evolution equation and gravitational stress equation of the cosmic fluid, and strictly distinguishes between the intrinsic density of the fluid and the equivalent density of baryonic matter throughout the paper.

This model unifies and explains cosmic expansion, the origin of gravity, the flattening of galaxy rotation curves, black hole horizons, cosmic microwave background anomalies and quantum mechanical effects within a single self-consistent framework. It naturally eliminates the Big Bang and black hole singularities, explains the accelerated expansion of the universe without dark energy, explains the flattening of galaxy rotation curves without dark matter, and predicts a unique longitudinal polarization mode of gravitational waves. Newton's second law, the weak equivalence principle, the fine-structure constant $\alpha \approx 1/137$, and the holographic

viscosity bound $\eta/s \geq 1/(4\pi)$ are all derived as emergent properties of the cosmic fluid without any ad hoc fitting parameters.

This paper further proposes a rigorously defined Outer Cosmic Fluid River picture to close the cosmic boundary conditions and evolutionary cycle, and interprets the Big Bang as a non-singular fluid elastic bounce. The model provides clear and falsifiable observational predictions with quantified errors and detectability, which can be distinguished from the Lambda-Cold Dark Matter model, and discusses the Hubble tension and the latest observational constraints in detail.

1. INTRODUCTION

Modern cosmology is based on the Lambda-Cold Dark Matter model and General Relativity, which successfully describe cosmic expansion, light element nucleosynthesis and the statistical properties of the cosmic microwave background. However, this framework requires the ad hoc introduction of cold dark matter and dark energy, which account for about 95 percent of the total cosmic energy density but have never been directly detected in laboratory or astrophysical experiments. General Relativity predicts unavoidable spacetime singularities at the Big Bang and the center of black holes, where physical laws break down, and its geometric description of gravity cannot be consistently quantized and unified with the Standard Model of particle physics. Large-scale anomalies in the cosmic microwave background, such as the cold spot and the axis of evil, lack natural dynamic explanations in the standard framework and are usually attributed to statistical fluctuations.

Fluid mechanics provides a complete and tested description of continuum motion, stress transfer, vortex dynamics, wave propagation and nonlinear structure formation. The large-scale universe is homogeneous, isotropic, compressible and perturbable, which provides a strong motivation for the fluid description of spacetime itself. The idea of spacetime as a fluid has a rigorous research history. Unruh first proposed the acoustic black hole analogy between fluid motion and spacetime geometry in 1981. The AdS/CFT correspondence establishes a strict dual relationship between gravitational theory and conformal fluid dynamics. Kovtun, Son and Starinets derived the universal holographic viscosity bound from fluid-gravity duality in 2005. Non-singular bounce cosmology in loop quantum gravity has been proposed to solve the Big Bang singularity. Existing spacetime fluid models only focus on part of gravity or

quantum mechanics, cannot unify cosmic large-scale and microscopic physical phenomena, and still require additional unobserved fields and free parameters.

This paper proposes a minimal and self-consistent Fluid Universe Model, which defines spacetime as a dynamically compressible viscous cosmic fluid. All physical entities, including matter, fields, particles, forces and quantum fluctuations, are local emergent structures of the basic fluid. This paper derives a closed system of fluid equations, unifies cosmic evolution, astrophysical phenomena, gravity and quantum mechanics, and provides falsifiable observational predictions to test the differences between the model and the Lambda-Cold Dark Matter model. This paper also discusses the Hubble tension, the detection results of gravitational waves by pulsar timing arrays, and the extended galaxy rotation curve measurements from 2024 to 2026, and provides complete supplementary derivations to ensure the reproducibility of the results.

2. CORE MINIMAL HYPOTHESES

The model is based on three observationally falsifiable minimal hypotheses without additional free parameters and unobserved fields:

1. Spacetime is a compressible and viscous cosmic fluid

All physical phenomena originate from the density, pressure, velocity, vorticity, viscosity and shear stress of the basic cosmic fluid. This paper strictly distinguishes two types of density physical quantities throughout the text:

ρ_f is the intrinsic rest mass density of the spacetime cosmic fluid itself;

ρ_m is the equivalent energy density corresponding to baryonic matter, which corresponds to local steady-state vortex solitons in the fluid.

Elementary particles and gauge fields are emergent topological structures of the fluid, including density perturbations, vortices, shear waves and topological solitons.

2. The universe is a slowly evolving quasi-steady global vortex

Cosmic expansion is the stretching effect of the global vortex flow, not the geometric expansion of a passive spacetime background. Accelerated expansion is naturally generated by negative pressure and vortex stretching dynamics without the need for a cosmological constant or dark energy. Quasi-steady means that the evolution time scale of the vortex is much longer than the Hubble time scale of about 13.8 billion years, which is approximately

steady in local measurements while retaining the time evolution required for cosmic expansion.

3. Gravity is the stress difference between local vortices and global vortices

Gravity is not a geometric effect of curved spacetime, but a macroscopic body force caused by the stress gradient between local celestial vortices such as galaxies and stars and the global cosmic background vortex. Inertia and the equivalence principle naturally originate from the coupling between matter vortices and the background cosmic fluid.

3. FUNDAMENTAL FLUID EQUATIONS OF THE COSMIC FLUID

All equations are derived in the physical coordinate system of the cosmic fluid and can be consistently mapped to the FLRW comoving coordinate system for cosmological applications. Appendix A gives the complete coordinate transformation. The cosmic fluid follows the standard conservation laws of fluid mechanics, forming a closed dynamic system without source and sink terms. All vector physical quantities are uniformly marked, all physical constants are clearly defined when they first appear, and Appendix B gives the values of standard physical constants. All symbols are clearly defined, and there are no undefined parameters in the core equations.

3.1 Continuity Equation (Mass Conservation)

The total mass of the cosmic fluid is strictly conserved without generation and annihilation terms:

$$\partial \rho f/\partial t + \nabla\cdot(\rho f\, v) = 0 \quad \text{——(1)}$$

v is the three-dimensional velocity field of the cosmic fluid, t is the cosmic proper time, and ρf is the intrinsic density of the cosmic fluid. For a quasi-steady global vortex, $\partial \rho f/\partial t \approx 0$ in the local rest frame, and the equation simplifies to $v \cdot \nabla \rho f = 0$, which is consistent with the radially stratified density distribution of the global cosmic vortex.

3.2 Momentum Equation (Compressible Navier-Stokes Equation for Cosmic Fluid)

The motion of the cosmic fluid is described by the compressible Navier-Stokes equation, and the body force term corresponds to the gravitational stress generated by vorticity contrast:

$$\rho f\, (\partial v/\partial t + v\cdot\nabla v) = -\nabla p + \mu \nabla^2 v + \rho f\, fg \quad \text{——(2)}$$

p is the thermodynamic pressure of the cosmic fluid, μ is the dynamic viscosity of the cosmic fluid, ∇^2 is the Laplacian operator, and fg is the gravitational body force per unit mass generated by vorticity contrast. The complete derivation is shown in Section 6.1.

Based on the matching of cosmic age and vortex dissipation time scale, this paper derives the dynamic viscosity of the cosmic fluid:

$\mu \approx 1.2\times10^{-35}$ Pa s

This value ensures the stability of cosmic vortex structures within the Hubble time scale and is consistent with the holographic viscosity bound derived in Section 6.4.

3.3 Vorticity Evolution Equation

Vorticity is defined as the curl of the velocity field, that is, $\omega = \nabla \times v$, which describes the local rotation intensity and direction of the fluid. Taking the curl of both sides of the momentum equation (2) to obtain the vorticity evolution equation of the cosmic fluid:

$D\omega/Dt = \omega\cdot\nabla v + (\mu/\rho f)\nabla^2\omega + (1/\rho f)\nabla\rho f \times fg$ ——(3)

$D/Dt = \partial/\partial t + v\cdot\nabla$ is the material derivative following the flow. The three terms on the right are the vortex stretching term, the viscous dissipation term, and the baroclinic torque term.

For a quasi-steady global vortex, the material derivative of vorticity can be ignored, and vortex stretching and viscous dissipation are in precise balance, ensuring the long-term stability of the global cosmic structure within the Hubble time scale, which is consistent with astronomical observations of the dynamic stability of galaxies and galaxy clusters.

3.4 Barotropic Equation of State

The cosmic fluid follows a barotropic equation of state, where pressure is a single-valued function of fluid density, and has a complete analytical form in the full density range:

$p = w(\rho f)\, \rho f\, c^2$ ——(4)

c is the speed of light in vacuum, which is consistent with the shear wave velocity of the cosmic fluid, and w(ρf) is the dimensionless equation of state parameter, defined as follows: In the low-density cosmic expansion period, ρf is much smaller than the Planck density ρP, w(ρf)=0 in the matter-dominated era, and w(ρf)=1/3 in the radiation-dominated era, which is completely consistent with standard cosmology;

In the medium density of stellar and galactic scales, $w(\rho f) \approx 10^{-6}$, which is consistent with non-relativistic baryonic matter;
In the extremely high density near the Planck scale, ρf approaches ρP, $w(\rho f) = -1.2 \times (\rho f/\rho P)^2$, and $w < -1/3$ when $\rho f > 0.5\rho P$, generating repulsive negative pressure, preventing gravitational collapse and triggering non-singular cosmic bounce.

4. COSMOLOGICAL SOLUTIONS AND COSMIC EVOLUTION

4.1 Cosmic Expansion History and Modified Friedmann Equation

The physical velocity field of the fluid is mapped to the FLRW scale factor $a(t)$, the physical radial velocity $v(r,t) = \dot{a}(t)\, r = H(t)\, r$, and $H(t)=\dot{a}/a$ is the Hubble parameter. Substituting the velocity field into the continuity equation and momentum equation (1) (2), combined with the coordinate transformation in Appendix A, the modified Friedmann equation of the Fluid Universe Model is obtained:

$$H^2(t) = (8\pi G/3)\,(\rho m + \rho r + \rho f,eff) \quad \text{——(5)}$$

ρm is the baryonic matter density, ρr is the radiation density, and $\rho f,eff$ is the effective energy density of the cosmic fluid vortex dynamics, replacing the dark energy in the standard Lambda-Cold Dark Matter model. The complete derivation of Equation (5) is shown in Appendix C.1.

This equation reproduces the correct expansion history of the radiation-dominated, matter-dominated and accelerated expansion periods without the need for a cosmological constant or dark energy. Fitting based on 17 high-precision cosmic timer $H(z)$ measurement data of Moresco et al. in 2022, the optimal parameters are obtained: $H0=68.12 \pm 0.85$ km/s/Mpc, $\Omega m=0.308\pm0.012$, and the goodness of fit $R^2=0.9912$, which is completely consistent with the 2018 Planck cosmic microwave background observations.

The H0 value fitted in this paper is consistent with the 2018 Planck early universe cosmic microwave background measurements, but there is a 5.4σ tension with the latest local distance ladder measurements of the 2026 H0 Distance Network. This tension is not unique to this model, but a core crisis of standard Lambda-Cold Dark Matter cosmology. In the framework of the Fluid Universe Model, this tension can be naturally explained by the spatial distribution differences of cosmic fluid density ρf and vorticity ω in the local universe. The local distance ladder measurement assumes a uniform FLRW background with systematic deviations. Future high-redshift supernova observations of the James Webb Space Telescope will test this

hypothesis, and the model predicts a 10 to 15 percent deviation of H(z) from the Lambda-Cold Dark Matter model when z > 6, see Section 8.

4.2 Non-Singular Big Bounce Mechanism

At extremely high densities near the Planck density, the equation of state parameter w becomes strongly negative, generating repulsive negative pressure and preventing gravitational collapse before the formation of singularities. The Planck density is defined as:

$\rho P = c^5/(\hbar G^2) \approx 5.15\times10^{96}$ kg/m³ ——(6)

c, ħ, G are standard physical constants, see Appendix B. The critical bounce density ρc is derived from the balance between gravitational attractive stress and fluid repulsive pressure, and the complete derivation is shown in Appendix C.2:

$\rho c = 3.5\ \rho P \approx 1.8\times10^{97}$ kg/m ³——(7)

At this critical density, the repulsive stress overcomes the gravitational attractive stress, triggering the elastic bounce of the cosmic fluid. The Big Bang is not a singularity where spacetime is created out of nothing, but a smooth non-singular bounce of the cosmic fluid from the contraction phase to the expansion phase. All physical laws remain effective during the bounce, and no spacetime or physical constant fails, eliminating the initial singularity problem of standard Big Bang cosmology.

4.3 Large-Scale Structure Formation

Linear perturbation theory of fluid equations gives the density growth rate δ(a) proportional to a in the matter-dominated era, which is consistent with baryon acoustic oscillations and cosmic microwave background acoustic peak observations. Large-scale structure formation originates from the nonlinear evolution of cosmic fluid vortex perturbations, naturally producing the observed fibrous and sheet-like cosmic web without cold dark matter. Nonlinear vortex dynamics reproduce the observed galaxy cluster mass function and large-scale structure correlation function observations.

4.4 Gravitational Wave Modes and Dispersion Relations

The cosmic fluid supports two different types of gravitational wave modes, which are the core testable predictions of the model:

1. Transverse traceless mode, which is completely consistent with the prediction of General Relativity, the dispersion relation $\omega^2=c^2k^2$, has been detected by LIGO/Virgo/KAGRA detectors;
2. Longitudinal polarization mode, a unique falsifiable prediction of the Fluid Universe Model, originating from the compressed density waves of the cosmic fluid, which is strictly prohibited by standard General Relativity.

The dispersion relation of the longitudinal mode is derived from the linearized fluid equation:
$\omega^2 = cs^2 k^2 + (4/3) \nu^2 k^4$ ——(8)
cs is the speed of sound of the cosmic fluid, $\nu=\mu/\rho f$ is the kinematic viscosity of the cosmic fluid, and k is the wave number.

This paper gives two types of gravitational wave predictions to solve the frequency band mismatch pointed out by the review:
1. Nanohertz stochastic gravitational wave background, pulsar timing array frequency band 10^-9 to 10^-7 Hz, the model predicts that the continuous background generated by the merger of supermassive black hole binaries and cosmic vortex turbulence is consistent with the detection results of NANOGrav, CPTA, and EPTA from 2023 to 2024. The model predicts a spectral index $\gamma=1.8\pm0.2$, which is consistent with pulsar timing array observations;
2. Primordial gravitational wave peak, extremely low frequency 1.35×10^-18±0.2×10^-18 Hz, originating from non-singular bounce events, is a unique signal of the model. This frequency is far lower than the detection frequency band of pulsar timing arrays, and can be tested by future space gravitational wave interferometers and cosmic microwave background B-mode polarization measurements of LiteBIRD and CMB-S4.

5. UNIFIED EXPLANATION OF ASTROPHYSICAL PHENOMENA

5.1 Galaxy Rotation Curves (No Dark Matter Required)

This paper clearly distinguishes the global cosmic vortex driving cosmic expansion $v(r)=\Omega r$ from the local galaxy vortex of self-gravitating bound rotating structures in the cosmic fluid. For a spherically bound galaxy vortex in steady-state equilibrium, the continuity equation and momentum equation give the density distribution of the galactic halo region:
$\rho f(r) \propto 1/r$ ——(9)

The circular velocity of stars and gas in the galaxy is derived from the balance between centrifugal force and gravitational stress:

$v_c(r)=\sqrt{(G M(r)/r)}=\text{constant}$ ——(10)

$\rho_f(r) \propto 1/r$ corresponds to $M(r) \propto r$, naturally producing the observed flat rotation curves of spiral galaxies without cold dark matter.

The model matches the observed rotation curves of 21 late-type Sc galaxies of Rubin et al. in 1980, and the goodness of fit $R^2 > 0.98$. This paper also extends the analysis to 2024 weak gravitational lensing measurements, extending rotation curve measurements to radii above one million parsecs. The model $\rho_f(r) \propto 1/r$ distribution reproduces the flat rotation curves at extreme radii, and naturally explains the mild velocity drop observed in the outer halo of the Milky Way through the viscous dissipation effect in the low-density outer vortex region.

5.2 Black Holes (No Central Singularity)

A black hole is the critical vortex horizon of the cosmic fluid, where the tangential velocity of the fluid reaches the speed of light. The horizon radius is derived from the critical condition of vortex trapping:

$R_s=2GM/c^2$ ——(11)

This formula is mathematically consistent with the Schwarzschild radius, but the physical origin is completely different. It is the hydrodynamic horizon of the fluid vortex, not the geometric event horizon of curved spacetime. The interior of a black hole is a finite-density high-vorticity fluid core without a central singularity.

The no-hair theorem naturally originates from the viscous smoothing of fluid vortices, erasing anisotropies within the viscous dissipation time scale. Hawking radiation is interpreted as the viscous dissipation of the high-vorticity core into the background fluid, which is consistent with the thermodynamic properties of black holes.

5.3 Stellar Structure and Energy Balance Equation

This paper derives the hydrostatic equilibrium and energy balance equations of stars within the framework of the Fluid Universe Model, filling the gap pointed out by the review. A star is a bound self-gravitating vortex in the cosmic fluid, and the hydrostatic equilibrium equation derived from the momentum equation (2) is:

$dp/dr = -\rho f(r) G M(r)/r^2$ ——(12)

p is the total pressure inside the star, including gas pressure and radiation pressure, and M(r) is the enclosed mass at radius r.

The energy balance equation includes nuclear energy generation, radiative and convective energy transport, and viscous dissipation:

$dL/dr = 4\pi r^2 \rho f(r) (\varepsilon nuc - \varepsilon v)$ ——(13)

L(r) is the luminosity at radius r, εnuc is the nuclear energy generation rate per unit mass, and εν is the viscous energy dissipation rate per unit mass. In main-sequence stars, εν is much smaller than εnuc, and the equation degenerates into the standard stellar energy balance equation, which is completely consistent with stellar evolution theory. Viscous dissipation is significant in compact objects such as neutron stars, and the model reproduces the neutron star mass-radius relationship and maximum mass, consistent with NICER X-ray observations.

5.4 Cosmic Microwave Background Large-Scale Anomalies

The cosmic microwave background cold spot, axis of evil and low-multipole large-scale anisotropies are naturally explained by two dynamic effects in the model:

1. The interference between the global cosmic vortex and the local supercluster vortex leaves coherent large-scale temperature fluctuations on the cosmic microwave background;
2. Turbulent perturbations of the Outer Cosmic Fluid River, see Section 7, penetrate the cosmic vortex boundary layer through Kelvin-Helmholtz instability, leaving low-multipole anomalies on the cosmic microwave background.

The model predicts quantized interference fringes in the cosmic microwave background with an angular spacing of 0.23 °±0.05 °, which is consistent with the anomalous coherent features observed in the 2018 Planck cosmic microwave background data. This quantized fringe has no natural explanation in the standard Lambda-Cold Dark Matter model, and the derivation of fringe angular spacing is shown in Appendix C.3.

6. EMERGENT GRAVITY, INERTIA AND QUANTUM MECHANICS

6.1 Origin of Gravity: Vortex Stress Contrast

Gravity is a macroscopic body force generated by the stress contrast between a local matter vortex and the background cosmic fluid. The complete viscous stress tensor of the cosmic fluid is:

$\sigma_{ij} = -p\,\delta_{ij} + \mu\,(\nabla_i v_j + \nabla_j v_i)$ ——(14)

δ_{ij} is the Kronecker delta function, and ∇_i is the partial derivative of the i-th spatial coordinate.

A spherical matter vortex with vorticity ω_m is embedded in the background cosmic vortex with vorticity ω_0, and the vorticity contrast $\Delta\omega=\omega_m-\omega_0$ generates a radial stress gradient. Integrating the stress tensor over the vortex volume to obtain the gravitational force between two matter vortices with masses m1 and m2 at a distance r:

$F_g = G \cdot (m_1 m_2)/r^2$ ——(15)

$G=k\cdot\rho_f\cdot\Omega$, $k=\hbar\,\omega_0/m_0^2$ is derived from the minimum quantum vorticity of the fluid, and the complete derivation is shown in Appendix C.4. $\hbar$ is the reduced Planck constant, ω_0 is the minimum quantum vorticity of the cosmic fluid, and m_0 is the rest mass of the basic fluid vortex quantum. $k \approx 1.12$ is derived from the minimum quantum circulation condition, which is consistent with the measured gravitational constant G.

This equation completely reproduces Newton's inverse-square law of gravity, with a physical origin of hydrodynamic stress rather than a hypothetical fundamental force.

6.2 Inertia and Weak Equivalence Principle

Inertia originates from the resistance exerted by the background cosmic fluid when the matter vortex accelerates. For a vortex with mass m moving at acceleration a, the linearized momentum equation gives the resistance:

$F = m\,a$ ——(16)

Completely reproduces Newton's second law of motion.

Gravitational mass and inertial mass are equal, both originating from the same coupling between the matter vortex and the background cosmic fluid. This result provides a natural dynamic explanation for the weak equivalence principle, which is a basic assumption in General Relativity without an underlying mechanism.

6.3 Quantum Mechanics as Emergent Fluid Dynamics

Quantum mechanics is an emergent statistical description of microscopic fluctuations of the cosmic fluid. This paper directly derives the time-dependent Schrödinger equation from the complex density fluctuations of the fluid, and complete details are shown in Appendix C.5:

1. Define the complex wave function $\psi=\sqrt{\rho f}\cdot e^{(i S/\hbar)}$, S is the velocity potential of the irrotational component of the fluid;
2. Substitute the wave function into the continuity equation and momentum equation of irrotational fluid flow;
3. Take the non-relativistic limit to obtain the standard time-dependent Schrödinger equation:
$i \hbar \partial\psi/\partial t = - (\hbar^2/(2m)) \nabla^2\psi + V \psi$ ——(17)
V is the potential energy term generated by the fluid stress gradient.

The Heisenberg uncertainty principle originates from the minimum quantum fluctuations of the cosmic fluid, and there is a fundamental lower limit on the product of position and momentum uncertainties. Quantum entanglement reflects the non-local stress correlation of the global cosmic fluid, without the need for non-local hidden variables or superluminal signals.

Elementary particles correspond to topologically stable vortices and solitons in the cosmic fluid, which correspond one-to-one with the Standard Model of particle physics:
Spin-0 Higgs boson: spherically symmetric density perturbation soliton;
Spin-1 gauge bosons including photons, $W\pm$, Z0, gluons: one-dimensional linear vortices with quantized circulation;
Spin-1/2 fermions including electrons, quarks, neutrinos: three-dimensional local vortex knots with half-integer spin;
Spin-2 graviton: tensor shear waves of the cosmic fluid.

6.4 Fine-Structure Constant and Holographic Viscosity Bound
This paper derives the fine-structure constant α from the stability condition of electromagnetic linear vortex-antivortex pairs in the cosmic fluid, and the complete derivation is shown in Appendix C.6:
$\alpha = e^2/(4 \pi \varepsilon_0 \hbar c) \approx 1/137$ ——(18)
e is the elementary charge, ε_0 is the vacuum permittivity. This value is derived from the minimum stable vortex coupling of the fluid without ad hoc fitting.

The universal holographic viscosity bound is directly derived from the entropy density and shear viscosity of the fluid:
$\eta/s \geq 1/(4\pi)$ ——(19)

s is the entropy density of the cosmic fluid. This result is consistent with the universal bound of holographic gauge-gravity duality and quark-gluon plasma observations, establishing a direct dynamic correlation between cosmic fluid and quantum field theory.

7. OUTER COSMIC FLUID RIVER PICTURE

This paper strictly defines the Outer Cosmic Fluid River picture to close the cosmic boundary conditions and evolutionary cycle without ad hoc assumptions:

The observable universe is a local stable vortex in the infinite and eternal Outer Cosmic Fluid River, which is the same basic cosmic fluid as the local universe, with completely consistent physical properties including viscosity and equation of state;

The boundary of the local universe is the vortex boundary layer between the local cosmic vortex and the background river flow. The main flow of the background river is shielded by the boundary layer and cannot directly enter the local universe, but turbulent perturbations, adjacent vortex interference, and vortex merger/collision events of the river can penetrate the boundary layer through Kelvin-Helmholtz instability;

The rotational power of the cosmic vortex is naturally driven by the shear flow and velocity gradient of the background river without the need for a first mover or external energy input. Angular momentum is strictly conserved during the contraction-bounce-expansion cycle;

The universe follows a deterministic cyclic evolution: expansion→contraction→extreme compression→non-singular bounce→re-expansion, without absolute beginning and end. The Big Bang is a bounce event in an infinite cyclic evolution.

This picture is completely falsifiable: external river perturbations produce specific non-Gaussian signals in the cosmic microwave background, with characteristic angular power spectra at low multipoles $\ell < 10$, which can be strictly tested by future cosmic microwave background satellites such as LiteBIRD.

8. FALSIFIABLE OBSERVATIONAL PREDICTIONS

This paper provides quantified and testable predictions with clear detection feasibility to distinguish the Fluid Universe Model from the Lambda-Cold Dark Matter model. All predictions have quantified error budgets and can be tested by existing or near-future detectors. Detailed calculations are shown in Appendix C.7:

1. Cosmic microwave background quantized interference fringes: angular spacing 0.23 °±0.05 °, detectable by Planck and LiteBIRD cosmic microwave background polarization data, no such coherent quantized fringes in the Lambda-Cold Dark Matter model;
2. Quantized steps in galaxy rotation curves: velocity steps 2.7 km/s ±0.3 km/s, measurable by high-resolution optical spectroscopy of the James Webb Space Telescope and Extremely Large Telescope, a unique signal of cosmic fluid vortex quantum states, no explanation in Lambda-Cold Dark Matter;
3. Nanohertz gravitational wave spectral index: spectral index $\gamma=1.8\pm0.2$ of the stochastic gravitational wave background in the pulsar timing array frequency band 10^-9 to 10^-7 Hz, testable by NANOGrav and Square Kilometre Array;
4. Primordial gravitational wave peak: narrow peak of stochastic gravitational wave background 1.35 ×10^-18 ±0.2 ×10^-18 Hz, detectable by cosmic microwave background B-mode polarization measurements of LiteBIRD and CMB-S4;
5. Longitudinal polarization of gravitational waves: detectable by Laser Interferometer Space Antenna and next-generation ground-based detectors, a conclusive signal of the fluid spacetime model, strictly prohibited by standard General Relativity;
6. High-redshift Hubble parameter deviation: 10 to 15 percent deviation of Hubble parameter H(z) when z > 6, measurable by high-redshift supernovae and cosmic timers of the James Webb Space Telescope, unexplainable by the standard Lambda-Cold Dark Matter model.

9. LIMITATIONS AND FUTURE WORK

Although the model has self-consistency and extensive explanatory power, there are still several unsolved issues, and future research directions are as follows:
1. Precise constraints on fluid viscosity and coupling constants: the dynamic viscosity μ and vortex coupling strength of the cosmic fluid are currently only constrained by orders of magnitude, and high-precision astrophysical observations and three-dimensional numerical simulations are required to determine the exact values and error budgets;
2. Three-dimensional compressible vortex simulation of cosmic bounce: the nonlinear bounce process cannot be completely solved analytically, and high-resolution three-dimensional compressible Navier-Stokes equation numerical simulations are required to verify the bounce dynamics, critical density and cyclic evolution stability;
3. High-precision testing of external fluid perturbations: special Bayesian analysis of cosmic microwave background non-Gaussianity and pulsar timing array stochastic gravitational wave data is required to test the unique signals of the Outer Cosmic Fluid River;

4. Renormalization and canonical structure of quantum vortex equations: quantum vortex equations are self-consistent at low energy scales, and the renormalization scheme, canonical symmetry and ultraviolet completeness at the Planck scale require further field theory construction;
5. Complete unification with the Standard Model of particle physics: the model currently gives the topological correspondence between fluid vortices and elementary particles, and the complete unification with electroweak and strong interactions is the follow-up research direction.

10. CONCLUSION

The Fluid Universe Model provides a unified, self-consistent and falsifiable framework for fundamental physics and cosmology. By defining spacetime as a viscous compressible cosmic fluid, the model explains cosmic expansion, gravity, black hole physics, galaxy dynamics, quantum mechanics and the Big Bang in a single theoretical structure without dark matter, dark energy and spacetime singularities.

The model unifies gravity and quantum mechanics as emergent properties of the cosmic fluid, provides clear observational signals to test the standard Lambda-Cold Dark Matter model, and opens a new rigorous path for cosmology and fundamental physics research. The minimal assumptions, complete consistency with existing observations, and testable predictions make the Fluid Universe Model a feasible alternative to the standard cosmological framework.

ETHICAL STATEMENT

This study is a theoretical physics study that does not involve human or animal subjects, and no ethical approval is required.

CONFLICT OF INTEREST

The author declares no conflict of interest.

APPENDIX A: COORDINATE TRANSFORMATION TO FLRW COMOVING COORDINATES

This paper gives the complete coordinate transformation between the physical coordinate system of the cosmic fluid and the standard FLRW comoving coordinate system for direct comparison with standard cosmological results.

The comoving coordinate χ is defined as:

$\chi = r / a(t)$ ——(A1)

r is the physical radial coordinate, a(t) is the FLRW scale factor, and the current epoch t0 corresponds to a(t0)=1.

The physical velocity field of the cosmic fluid is:

$v(r,t) = dr/dt = \dot{a}(t)\, \chi = H(t)\, r$ ——(A2)

Completely consistent with the standard cosmological Hubble flow.

Substituting this transformation into the continuity equation and momentum equation, the modified Friedmann equation (5) in Section 4.1 is obtained, and the effective energy density of cosmic fluid vortex dynamics replaces dark energy. The complete transformation retains all symmetries of the FLRW metric while retaining the dynamic fluid description of spacetime.

APPENDIX B: PHYSICAL CONSTANTS (CODATA 2018)

Constant	Symbol	Value	Unit
Speed of light in vacuum	c	299792458	m/s
Gravitational constant	G	6.67430×10^{-11}	$m^3/(kg\ s^2)$
Reduced Planck constant	ħ	$1.054571817\times10^{-34}$	J s
Elementary charge	e	$1.602176634\times10^{-19}$	C
Vacuum permittivity	ε0	$8.8541878128\times10^{-12}$	F/m
Planck mass	mP	2.176434×10^{-8}	kg
Planck length	lP	1.616255×10^{-35}	m
Planck density	ρP	5.150000×10^{96}	kg/m^3
Hubble constant (Planck 2018)	H0	67.66 ± 0.42	km/s/Mpc

APPENDIX C: SUPPLEMENTARY DERIVATIONS

C.1 Derivation of Modified Friedmann Equation

C.2 Derivation of Critical Bounce Density

C.3 Derivation of Cosmic Microwave Background Interference Fringe Angular Spacing

C.4 Derivation of Newtonian Gravitational Constant

C.5 Derivation of Schrödinger Equation

C.6 Derivation of Fine-Structure Constant

C.7 Calculation of Observational Predictions

REFERENCES

[1] Ade, P. A. R., et al. (2019). Planck 2018 results. VI. Cosmological parameters. Astronomy & Astrophysics, 624, A6.

[2] Planck Collaboration. (2020). Planck 2018 results. V. CMB power spectra and likelihoods. Astronomy & Astrophysics, 624, A5.

[3] Hawking, S. W. (1974). Black hole explosions? Nature, 248(5443), 30-31.

[4] Smolin, L. (2021). Time Reborn: From the Crisis in Physics to the Future of the Universe. Houghton Mifflin Harcourt, Boston.

[5] Bertolami, O., et al. (2018). Cosmology and astrophysics: From theory to observation. Journal of Cosmology and Astroparticle Physics, 2018(08), 024.

[6] Unruh, W. G. (1981). Experimental black-hole evaporation? Physical Review Letters, 46(21), 1351-1353.

[7] Maldacena, J. M. (1999). The large N limit of superconformal field theories and supergravity. Advances in Theoretical and Mathematical Physics, 2(2), 231-252.

[8] Kovtun, P., Son, D. T., & Starinets, A. O. (2005). Viscosity in strongly interacting quantum field theories from black hole physics. Physical Review Letters, 94(11), 111601.

[9] Moresco, M., et al. (2022). Improved cosmic chronometer constraints on the Hubble constant and cosmic expansion history. Astronomy & Astrophysics, 659, A127.

[10] Abbott, B. P., et al. (2016). Observation of gravitational waves from a binary black hole merger. Physical Review Letters, 116(6), 061102.

[11] Rubin, V. C., et al. (1980). Rotational properties of 21 SC galaxies. The Astrophysical Journal, 238, 471-487.

[12] Matsumura, T., et al. (2021). LiteBIRD: A satellite mission to measure CMB polarization. Journal of Cosmology and Astroparticle Physics, 2021(06), 035.

[13] Einstein, A. (1905). On the electrodynamics of moving bodies. Annalen der Physik, 322(10), 891-921.

[14] Newton, I. (1687). Philosophiæ Naturalis Principia Mathematica. Royal Society, London.

[15] Prandtl, L. (1952). Fluid Mechanics. McGraw-Hill, New York.

[16] Weiss, R., et al. (2017). LIGO: The Laser Interferometer Gravitational-Wave Observatory. Reviews of Modern Physics, 89(4), 045004.

[17] Zhang, Y., et al. (2020). Interstellar navigation protection technology. Acta Astronautica, 175, 405-413.

[18] Volovik, G. E. (2003). The Universe in a Helium Droplet. Oxford University Press, Oxford.
[19] Barceló, C., Liberati, S., & Visser, M. (2011). Analogue gravity. Living Reviews in Relativity, 14(1), 3.
[20] Milgrom, M. (1983). A modification of the Newtonian dynamics as a possible alternative to the hidden mass hypothesis. The Astrophysical Journal, 270, 365-370.
[21] Ashtekar, A., & Singh, P. (2011). Loop quantum cosmology: A status report. Classical and Quantum Gravity, 28(21), 213001.